Loop Invariants and the Correctness Proof

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1 Loop Invariant ?

Loop invariant .

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1. **Initialization**
   loop invariant .
2. **Maintenance**
   invariant , .
3. **Termination**
   , invariant .

2 Insertion Sort Loop Invariant

Insertion Sort `for` ,
`A[1 ... j-1]` , .

2.1 Initialization

`j = 2` , `A[1 ... j-1]` `A[1]` .
, invariant .

2.2 Maintenance

`A[j]`
(`A[j-1]`, `A[j-2]`, …) .
`A[j]` `A[1 ... j]` .

`A[1 ... j-1]` .

2.3 Termination

`for` `j = n+1` .
invariant `A[1 ... n]` .

, **Insertion Sort** .